VF5 caustics: before / after

What it is: the moving net of light that the rippled water surface focuses onto sand, rock tops and wood.
How: one small pattern made once, sampled twice drifting apart; only on surfaces facing up, under water.
Sharp and strong under spot lights (reef), softer under LED bars and tubes. Off for dry tanks and no-fixture.
It redistributes light (brighter lines, slightly darker cells); the tank's overall brightness does not change.
Busy gravel (nano) hides it, as real gravel does; pale sand (planted, reef) shows it most.

Cost per frame (alternating A/B frames, GPU): planted +0.12 ms, nano +0.16 ms, reef +0.14 ms. Draw calls unchanged.
The bigger cost: it moves, so the tank redraws 20 times a second while you just look at it
(today the app draws only when something changes). That is about 20-25 % GPU busy at rest.
If that is too much, the option is still caustics (no motion, zero cost at rest).

Pictures: front, substrate close-up, rock close-up, for each of the three reference tanks.
Left = now (ACES), right = with caustics. Build on origin/main cf66a31e + VF5.

Planted 4 ft (LED bar) - front view. LEFT: now (ACES) RIGHT: with caustics

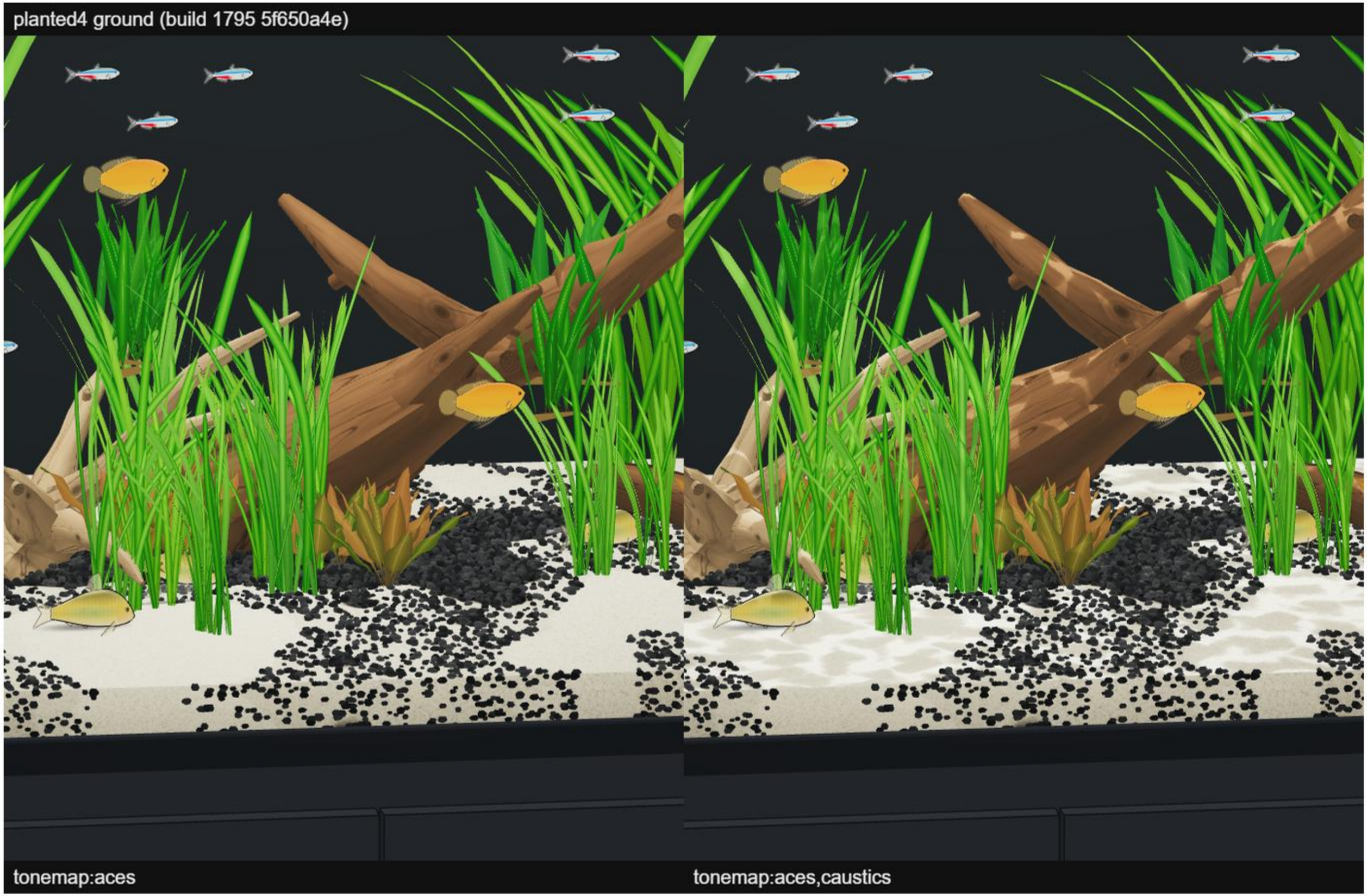
planted4 front (build 1795 5f650a4e)



tonemap:aces

tonemap:aces,caustics

Planted 4 ft (LED bar) - ground view. LEFT: now (ACES) RIGHT: with caustics



Planted 4 ft (LED bar) - rock view. LEFT: now (ACES) RIGHT: with caustics



Rimless nano (LED bar) - front view. LEFT: now (ACES) RIGHT: with caustics

nano front (build 1795 5f650a4e)

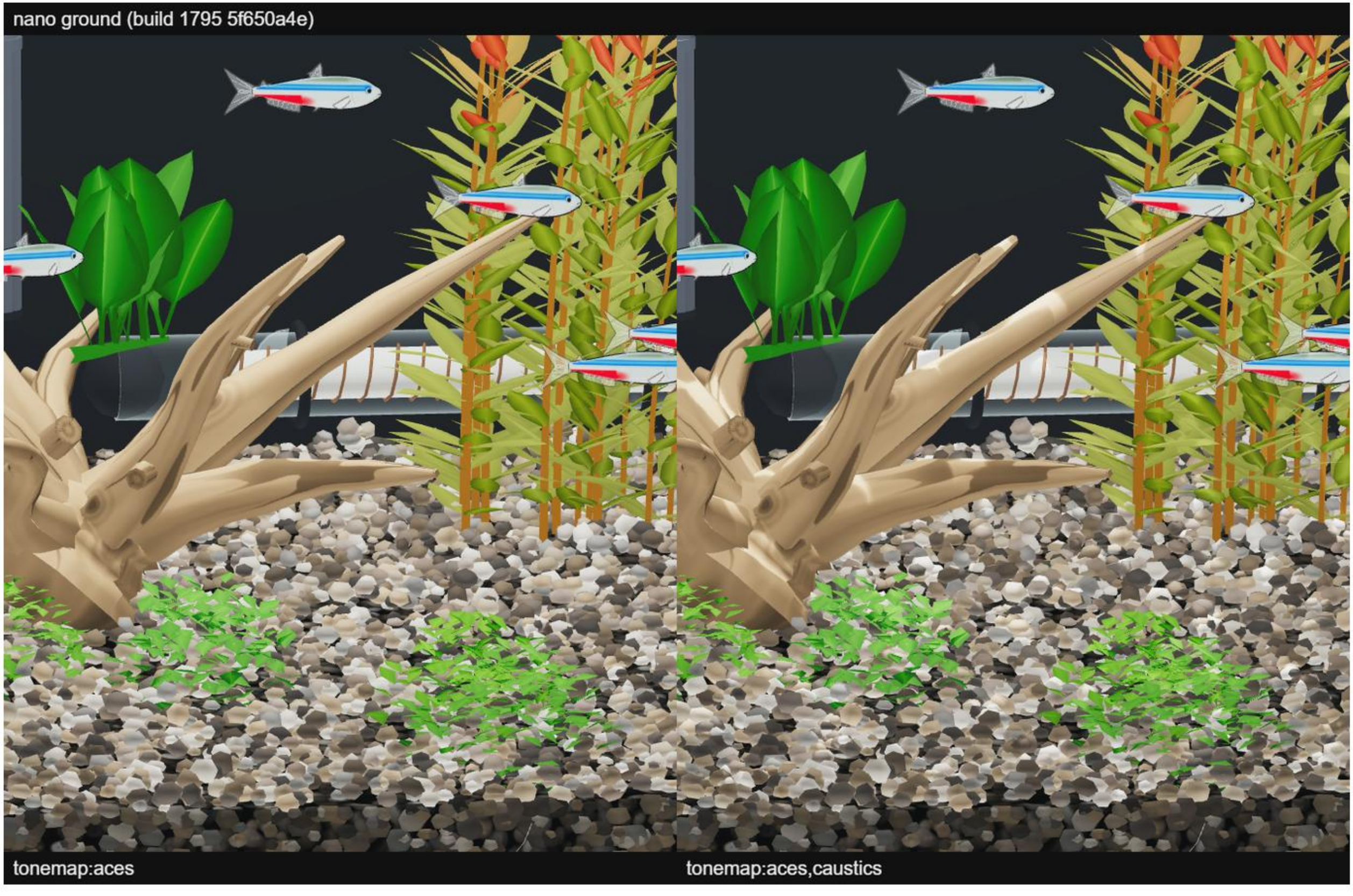


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Rimless nano (LED bar) - ground view. LEFT: now (ACES) RIGHT: with caustics



Rimless nano (LED bar) - rock view. LEFT: now (ACES) RIGHT: with caustics



Reef bow-front (spots) - front view. LEFT: now (ACES) RIGHT: with caustics



Reef bow-front (spots) - ground view. LEFT: now (ACES) RIGHT: with caustics



Reef bow-front (spots) - rock view. LEFT: now (ACES) RIGHT: with caustics

